

Homework_6: PDA

Question_1:

Construct PDA for $L = a^n b^m c^k : n, m, k \geq 0$ and $n=m$ or $n=k$

$(Q, \Sigma, \Gamma, \delta, q_0, \$, F)$

Transition function

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

$\delta(q_0, \epsilon, \epsilon) = \{(q_1, \$)\}$

$\Sigma = \{a, b, c\}$

$\delta(q_1, a, \epsilon) = \{(q_1, a)\}$

$\Gamma = \{a, b, c, \$ \}$

$\delta(q_1, \epsilon, \epsilon) = \{(q_2, \epsilon)\}$

Initial state q_0

$\delta(q_1, \epsilon, \epsilon) = \{(q_4, \epsilon)\}$

Stack start symbol $\$$

$\delta(q_2, b, a) = \{(q_2, \epsilon)\}$

$F = \{q_6\}$

$\delta(q_2, \epsilon, \epsilon) = \{(q_3, \epsilon)\}$

$\delta(q_3, c, \epsilon) = \{(q_3, \epsilon)\}$

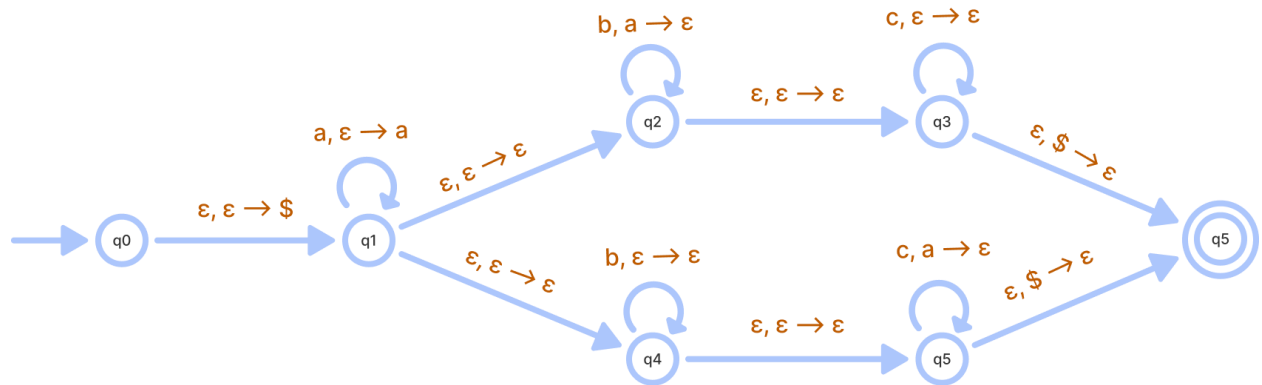
$\delta(q_3, \epsilon, \$) = \{(q_6, \epsilon)\}$

$\delta(q_4, b, \epsilon) = \{(q_4, \epsilon)\}$

$\delta(q_4, \epsilon, \epsilon) = \{(q_5, \epsilon)\}$

$\delta(q_5, c, a) = \{(q_5, \epsilon)\}$

$\delta(q_5, \epsilon, \$) = \{(q_6, \epsilon)\}$



Question_2:

Construct PDA for grammar

$S \rightarrow aBc \mid ab$

$B \rightarrow SB \mid \epsilon$

